JOSEPH MANNING ELLIS

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EDUCATION

University of Southern California — Graduated May 2026
Master of Science
Applied Data Science | Viterbi School of Engineering

Chapman University — Graduated May 2024
Bachelor of Science
Data Analytics | Fowler School of Engineering
Minor in Business Administration

TECHNICAL SKILLS

Programming: TypeScript | Python | SQL | R | C++ | React | Next.js | Node.js | Cursor | Claude Code
AI / LLM Systems: Prompt Engineering | RAG | LLM Orchestration | Multi-Agent Systems | Evaluation & Guardrails
Machine Learning: PyTorch | TensorFlow | Pandas | NumPy | NLP | Computer Vision | Deep Learning | LoRA Fine-Tuning
Data & Infrastructure: PostgreSQL | Supabase | Redis | GCP | AWS | Docker | Trigger.dev | REST APIs | OAuth 2.0

WORK EXPERIENCE

AI Software Engineer - Buildy — January 2026 – Present
Marketing Platform: Organic Social, Paid Ads & AI Media
- Built the Facebook, LinkedIn, X, and TikTok publishing integrations behind a single publish path that handles four different auth models and media constraints.
- Built the Google Ads integration from zero: OAuth, conversion tracking, tag injection, Search campaigns, and Demand Gen across video, image, and carousel, shipped behind staged launch gates.
- Shipped an AI media library that generates images and video across multiple model providers, with native credit metering and publishing to social platforms.
- Rebuilt AI content generation around one subagent with per-platform skills, in place of a chain of per-platform LLM generators, raising user approval of generated posts.
- Engineered the correctness layer behind campaign publishing and budgets: multi-currency handling and idempotent writes that reconcile against remote state.

Course Hub
- Built Course Hub, a product that takes a creator from a topic to a sellable course: AI-authored lessons, offers and pricing, publishing, narration with voice selection, and an AI tutor.
- Designed the preview/publish split for course content so creator edits reach learners only on publish, matching how project code already shipped.
- Shipped guest checkout where an account is created after payment and only on proof of payment, with creator payouts going directly to their own Stripe account.

Machine Learning Engineer - PushButton.AI — January 2024 – December 2025
- Built LLM backend services on Docker and Redis for an automated course-generation product, reducing API cost while evaluating open-source models against hosted ones.

Co-Founder | Backend Engineer - Ovrsee — April 2024 – October 2024
- Developed API ingestion workflows using Google Cloud Functions to support OAuth-based access to external platforms, including encrypted credential storage.
- Built automated token refresh workflows to maintain continuous API access across user accounts, preventing service interruptions and reducing authentication-related failures.

TECHNICAL PROJECTS

Multi-Agent LLM Simulation & Evaluation Framework — 2025
- Built a configurable multi-agent LLM simulation framework modeling social deduction dynamics (deception, persuasion, theory-of-mind) with structured observation/action loops and role-based state transitions
- Designed a config-driven experiment and evaluation pipeline benchmarking agent behavior across 499 games and 5 configurations. LoRA fine-tuning raised persuasion 321% and consensus from 45% to 80%.
- Developed modular agent architecture with JSON-constrained actions, memory/reflection modules, and moderation guardrails, enabling scalable experimentation on multi-agent coordination and decision-making

HONORS

Magna Cum Laude | Program Honors — Chapman University
Provost's List (7 Semesters) | Dean's List (1 Semester) — Chapman University
Excellence in Computing Studies — Association for Computing Machinery